EXECUTIVE TECHNICAL REPORT

Customer Churn & Retention Analytics

1. Executive Summary

This project presents an end-to-end data analytics and predictive machine learning solution designed to evaluate customer turnover, uncover underlying behavioral churn triggers, and deliver actionable retention strategies for executive leadership. By integrating a high-performance **Random Forest Classifier** trained on preprocessed customer data with an interactive **Power BI Executive Dashboard**, business stakeholders can proactively identify high-risk customer segments and minimize revenue leakage.

2. Tools & Technologies Demonstrated

Domain	Tools / Libraries / Languages	Application & Implementation Detail
Database & Data Pipeline	`SQL` (MySQL / Relational DB)	Data extraction, relational querying, structured dataset staging, and database management.
Data Preprocessing & EDA	`Python`, `Pandas`, `NumPy`	Data cleaning, missing value handling, structural transformations, and feature encoding.
Machine Learning	`Scikit-Learn` (`RandomForestClassifier`)	Model training, accuracy score (99.81%), ROC AUC evaluation, and probabilistic scoring.
Business Intelligence	`Power BI Desktop`, `DAX`	Custom DAX KPI measures, visual grid modeling, dynamic slicers, and interactive dashboarding.
Development & VCS	`Git`, `GitHub`, `VS Code`, `Markdown`	Version control, repository structure management, and professional project documentation.

TOTAL CUSTOMERS	TOTAL CHURN	CHURN RATE	HIGH-RISK USERS
64.37K	30K	47.37%	30K

3. Business Problem & Objectives

Customer retention is a critical metric directly tied to long-term profitability and customer lifetime value (LTV). Unplanned customer churn increases operational acquisition costs and degrades market share.

- **Primary Objective:** Build an enterprise-grade analytics pipeline that cleans baseline customer data, builds an accurate machine learning prediction model, and delivers dynamic visual insights via Power BI.
- **Key Business Goals:**
  1. Measure baseline operational KPIs including total customers (64.37K), total churn (30K), and overall churn rate (47.37%).
  2. Pinpoint behavioral patterns associated with account cancellation, such as support ticket volumes and contract durations.
  3. Deploy predictive risk scoring (`Churn_Probability` and `Predicted_Churn_Risk`) to empower retention teams.

4. Technical Architecture & Methodology

4.1 Data Cleaning & Preprocessing

Ingested baseline customer records (`customer_churn_cleaned.csv`) containing demographic attributes, usage metrics, support interaction logs, contract structures, and financial variables. Enforced dataset integrity across all records by performing full-dataset transformations without row truncation.

4.2 Machine Learning Pipeline Development

Selected and trained a **Random Forest Classifier** to accurately classify customer churn risk. Generated continuous probabilistic scoring (`Churn_Probability`) and categorical risk assignments (`Predicted_Churn_Risk` : Low Risk vs. High Risk). Exported the enriched dataset (`customer_churn_with_ml.csv`) to serve as the unified underlying data source for business intelligence reporting.

5. Key Performance Indicators & DAX Metrics

Metric Name	DAX Expression	Value / Significance
Total Customers	`COUNT(customer_churn[CustomerID])`	**64.37K** total base volume
Total Churn	`SUM(customer_churn[Churn])`	**30K** absolute churn loss
Churn Rate	`DIVIDE([Total Churn], [Total Customers], 0)`	**47.37%** relative loss percentage
High Risk Customers	`CALCULATE([Total Customers], Risk = "High Risk")`	**30K** ML-flagged intervention accounts

6. Machine Learning Model Benchmarks

- **Algorithm Architecture:** Random Forest Classifier
- **Model Accuracy Score: 99.81%**
- **ROC AUC Score: 1.000**
- **Key Outputs:** `Churn_Probability` and categorical `Predicted_Churn_Risk` flags.

7. Power BI Dashboard Visual Architecture & Findings

- **Executive KPI Header (Top Banner):** Displays real-time metrics for Total Customers, Total Churn, Churn Rate, and High-Risk count.
- **Interactive Slicers (Top Right):** Enables dynamic slicing across `Gender` and `Subscription Type` (Basic, Premium, Standard).
- **Support Call Dynamics (Column Chart):** Highlights a strong positive correlation between high support calls and increased customer churn volume.
- **Contract Tenure Breakdown (Bar Chart):** Compares churn rates across Monthly, Quarterly, and Annual contracts, revealing that Monthly subscribers exhibit the highest vulnerability.
- **ML Risk Segmentation (Donut Chart):** Visually breaks down customer distribution by machine learning risk levels (`Count of CustomerID by Predicted_Churn_Risk`).

8. Strategic Recommendations

1. **Automated Support Alerts:** Trigger proactive outreach for customers logging >3 support calls.
2. **Long-Term Contract Incentives:** Offer conversion discounts to migrate Monthly contract users to Annual plans.
3. **Targeted Interventions:** Focus retention campaigns exclusively on the 30K ML-flagged High-Risk accounts.

8. Dataset Architecture & Data Pipeline

The analysis utilizes a full-spectrum customer behavioral dataset (`customer_churn_cleaned.csv`) enriched via Python feature engineering into `customer_churn_with_ml.csv` .

Column Name	Data Type	Description / Source
`CustomerID`	String / Text	Unique customer identifier (Primary Key)
`Age` , `Gender`	Integer / Categorical	Demographic profiles (Male, Female)
`Tenure` , `Usage Frequency`	Integer	Product engagement and account tenure duration
`Support Calls`	Integer	Total customer support service tickets logged
`Contract Length`	Categorical	Subscription commitment (Monthly, Quarterly, Annual)
`Subscription Type`	Categorical	Tier level (Basic, Standard, Premium)
`Churn`	Binary (0 / 1)	Target ground truth label (1 = Churned, 0 = Retained)
`Churn_Probability`	Float [0.0 - 1.0]	Continuous predictive score generated by ML model
`Predicted_Churn_Risk`	Categorical	Machine learning risk group assignment (High Risk vs. Low Risk)

9. Python Machine Learning Implementation (`train_churn_model.py`)

The model was built using **Scikit-Learn**'s `RandomForestClassifier` . Below is the operational workflow script used to train, evaluate, and append probabilities:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, roc_auc_score

# 1. Ingest Data
df = pd.read_csv('customer_churn_cleaned.csv')

# 2. Features & Target Setup
X = pd.get_dummies(df.drop(columns=['CustomerID', 'Churn']), drop_first=True)
y = df['Churn']

# 3. Train Test Split & Model Fitting
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# 4. Evaluation Metrics
preds = model.predict(X_test)
probs = model.predict_proba(X)[:, 1]
print(f"Accuracy: {accuracy_score(y_test, preds):.4f}") # 99.81%
print(f"ROC AUC: {roc_auc_score(y, probs):.4f}")        # 1.000

# 5. Export Enriched Dataset
df['Churn_Probability'] = probs
df['Predicted_Churn_Risk'] = df['Churn_Probability'].apply(lambda x: 'High Risk' if x >= 0.5 else
'Low Risk')
df.to_csv('customer_churn_with_ml.csv', index=False)
```

10. Complete Power BI Visual Component Inventory

- **Header KPI Cards:** Formatted with custom DAX calculations displaying Total Customers (`64.37K`), Total Churn (`30K`), Churn Rate (`47.37%`), and ML High-Risk Count (`30K`).
- **Interactive Slicers:** Categorical selection filters for `Gender` (Male, Female) and `Subscription Type` (Basic, Premium, Standard) for instant cross-filtering.
- **Support Call Dynamics Chart:** Clustered column chart plotting `Support Calls` on the X-axis against `Sum of Churn` on the Y-axis.
- **Contract Type Analysis Chart:** Standard Clustered bar chart comparing churn volume across `Contract Length` (Monthly, Annual, Quarterly).
- **ML Risk Breakdown Donut Chart:** Ring visualization showing `Count of CustomerID` grouped by `Predicted_Churn_Risk` to highlight risk proportions.

11. GitHub Repository Deliverables Structure

The complete end-to-end repository is structured for production deployment and recruiter code-review:

- `customer_churn_cleaned.csv` — Cleaned baseline raw dataset.
- `customer_churn_with_ml.csv` — Enriched dataset with probabilities and risk classifications.
- `train_churn_model.py` — Python machine learning pipeline script.
- `Customer_Churn_Retention_Analysis.pbix` — Interactive Power BI visual dashboard.
- `requirements.txt` — Python environment dependencies (`pandas` , `scikit-learn` , `numpy`).
- `README.md` — Modern light-themed GitHub portfolio presentation.

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